

CLINICAL PRACTICE UPDATES

AGA Clinical Practice Update on Risk Stratification and Emerging Surveillance Strategies for Hepatocellular Carcinoma: Expert Review



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DESCRIPTION: Hepatocellular carcinoma (HCC) is a leading cause of cancer-related death in patients with cirrhosis. Although advancements in surgical, locoregional, and systemic therapies have improved outcomes for patients with early-stage HCC, most cases are diagnosed at later stages due to both the limitations of currently recommended surveillance modalities and their underuse, with fewer than 1 in 4 patients with cirrhosis receiving surveillance for this tumor. Current guidelines recommend semiannual surveillance with ultrasound and α -fetoprotein, but this strategy has suboptimal sensitivity, and alternative imaging techniques (eg, computed tomography and magnetic resonance imaging) have their own risks and challenges with implementation. HCC surveillance has established benefits in certain at-risk populations, including patients with cirrhosis (of any etiology) and selected patients with chronic hepatitis B virus. The burden of HCC is rising most rapidly among patients with nonviral liver diseases, including metabolic dysfunction-associated steatotic liver disease and alcohol-associated liver disease, among whom the annual incidence rate of HCC is significantly lower than was previously observed in patients with viral hepatitis. This underscores the need for better biomarker and risk-stratification tools to detect HCC earlier and improve surveillance efficiency, although integrating them into routine clinical practice presents a significant challenge. This American Gastroenterological Association (AGA) Clinical Practice Update aims to provide Best Practice Advice on emerging strategies for HCC risk stratification and surveillance. **METHODS:** This Expert Review was commissioned and approved by the AGA Institute Clinical Practice Updates Committee and the AGA Governing Board to provide timely guidance on this topic of high clinical importance to the AGA membership and underwent internal peer review by the Clinical Practice Updates Committee and external review procedures through the standard review process for *Gastroenterology*. These Best Practice Advice statements were drawn from a review of the published literature and from expert opinion. Because a systematic review was not performed, these Best Practice Advice statements do not carry formal ratings of the quality of available evidence or the strength of the presented advice.

BEST PRACTICE ADVICE STATEMENTS

BEST PRACTICE ADVICE 1: The best strategy for reducing hepatocellular carcinoma morbidity and mortality is to prevent cirrhosis. These strategies include vaccination and treatment of viral hepatitis (eg, hepatitis C virus and hepatitis B virus), recognizing and treating alcohol use disorder, managing metabolic syndrome, and addressing liver diseases at early stages. **BEST PRACTICE ADVICE 2:** The preferred current surveillance strategy for patients at risk of hepatocellular carcinoma is semiannual ultrasound and α -fetoprotein in combination. The benefits of surveillance include a higher likelihood of detecting early-stage hepatocellular carcinoma, access to curative therapy, and improved survival. **BEST PRACTICE ADVICE 3:** Among patients without cirrhosis, only a subset of those with chronic hepatitis B virus should undergo hepatocellular carcinoma surveillance; surveillance is not advised for those without cirrhosis from other etiologies due to the low annual incidence rate. **BEST PRACTICE ADVICE 4:** Although the benefits of hepatocellular carcinoma surveillance are well established, it is important for clinicians to consider the potential physical, psychological, and financial harms associated with the process. **BEST PRACTICE ADVICE 5:** Several novel blood-based biomarkers, such as GALAD, and radiologic biomarkers are undergoing clinical validation. Some are already commercially available, but evidence is insufficient to support their use in routine hepatocellular carcinoma surveillance. These assays should not replace guideline-recommended tests, although their accessibility and potential cost-effectiveness may support broader use once sufficiently validated. **BEST PRACTICE ADVICE 6:** Multicancer blood-based biomarker detection panels should not be used in the screening or surveillance of patients at risk for hepatocellular carcinoma. **BEST PRACTICE ADVICE 7:** Many hepatocellular carcinoma risk-stratification scores exist for patients with cirrhosis. However, few scores have undergone sufficient validation to support their use in clinical practice. **BEST PRACTICE ADVICE 8:** Among patients with chronic hepatitis B virus infection without cirrhosis, PAGE-B and REAL-B scores can stratify patients based on their future risk of hepatocellular carcinoma.

Keywords: Liver Cancer; Screening; Surveillance; Biomarker; Risk Scores; Prediction Models.

Hepatocellular carcinoma (HCC) is the third leading cause of cancer-related death worldwide and a major cause of mortality among patients with cirrhosis. Historically associated with a poor prognosis, survival rates have improved due to recent advancements in surgical, locoregional, and systemic therapies. As curative treatments are available for those diagnosed at an early stage, early detection is crucial for improving HCC survival.

Despite this, only 30%–40% of HCC cases are diagnosed at an early stage, largely due to limitations and underutilization of HCC surveillance. Societal guidelines recommend semiannual surveillance with ultrasound (US) and α -fetoprotein (AFP) for all patients at high risk for HCC; however, this strategy has suboptimal sensitivity in real-world practice. Uptake of HCC surveillance also remains far lower than for other cancers, with fewer than 1 in 4 patients with cirrhosis receiving consistent surveillance. At the same time, the epidemiology of cirrhosis and HCC in the United States is shifting from predominantly viral hepatitis C (HCV)-related disease toward nonviral liver diseases, particularly metabolic dysfunction-associated steatotic liver disease (MASLD) and alcohol-associated liver disease.¹ Although these nonviral etiologies have lower annual HCC incidence rates, they represent rapidly growing populations, which increases the number needed to screen and further complicates effective surveillance efforts.

Given these challenges, there is a critical need for improved biomarkers and more precise risk-stratification tools to improve the efficiency of HCC early detection. Several promising blood- and imaging-based biomarkers are in various stages of clinical validation; however, integrating these tools into routine clinical practice remains a key challenge.

This review provides best practice guidance on risk stratification and surveillance strategies for HCC. We have developed Best Practice Advice statements to address 8 key clinical issues.

Best Practice Advice 1: The best strategy for reducing hepatocellular carcinoma morbidity and mortality is to prevent cirrhosis. These strategies include vaccination and treatment of viral hepatitis (eg, hepatitis C virus and hepatitis B virus), recognizing and treating alcohol use disorder, managing metabolic syndrome, and addressing liver diseases at early stages.

Early identification and treatment of chronic liver disease is essential for reducing HCC risk. Viral hepatitis, particularly hepatitis B virus (HBV) and HCV, remains a leading cause of cirrhosis and HCC worldwide. The lifetime HCC risk for individuals with HBV is estimated to be 10%–25%.² HBV-related HCC is relatively uncommon in the United States, owing to the success of universal vaccination programs³ and effective antiviral therapies. Early identification and treatment of chronic HBV infection are crucial, as nucleos(t)ide analogs (eg, tenofovir and entecavir) suppress viral replication, reduce liver inflammation, and

prevent progression to cirrhosis. Patients should also be screened for coinfection with other viruses (HCV, hepatitis D virus, and HIV), which significantly increases HCC risk compared with HBV monoinfection.^{4,5}

For decades, HCV has been the predominant etiology of HCC in the United States. Direct-acting antivirals are effective, well-tolerated oral medications that have revolutionized the management of HCV. Achieving HCV cure not only halts further liver disease progression, but can also reverse some of the damage caused by cirrhosis, lowering HCC risk.⁶ Expanding access to HCV testing and treatment, particularly in high-risk populations is essential to reducing the overall burden of HCC.⁷

MASLD, previously known as nonalcoholic fatty liver disease, is highly prevalent in the United States and, alarmingly, the most rapidly increasing etiology of HCC.⁸ It is closely linked to metabolic syndrome, obesity, and type 2 diabetes (all established, independent risk factors for HCC⁹). Lifestyle modifications, weight loss, a balanced diet, and regular exercise, provide the foundation for preventing and managing MASLD. New therapies such as resmetirom—an oral, liver-directed thyroid hormone receptor β -agonist—and glucagon-like peptide 1 receptor agonists (and possibly the many other promising drugs in the MASLD pipeline with various mechanisms of action¹⁰) may reduce HCC risk by means of reducing fibrosis associated with MASH, but further studies with long-term follow-up are needed to confirm this effect.

Alcohol-associated liver disease is emerging as an increasing cause of HCC.¹¹ Chronic heavy alcohol consumption induces liver inflammation, steatosis, and fibrosis, leading to cirrhosis and increased risk of developing HCC. Prevention efforts include behavioral interventions, pharmacologic treatments for alcohol use disorder (eg, naltrexone), and public health policies like alcohol taxes and awareness campaigns.

Best Practice Advice 2: The preferred current surveillance strategy for patients at risk of hepatocellular carcinoma is semiannual ultrasound and α -fetoprotein in combination. The benefits of surveillance include a higher likelihood of detecting early-stage hepatocellular carcinoma, access to curative therapy, and improved survival.

HCC surveillance should be performed in at-risk individuals, particularly those with cirrhosis of any etiology and selected patients with chronic HBV infection, even in the absence of cirrhosis (Table 1).¹² Surveillance for patients with Child-Turcotte-Pugh class A or B cirrhosis is beneficial. In contrast, patients with Child-Turcotte-Pugh class C cirrhosis who are not candidates for liver

Abbreviations used in this paper: AFP, α -fetoprotein; HBV, hepatitis B virus; HCC, hepatocellular carcinoma; HCV, hepatitis C virus; MASLD, metabolic dysfunction-associated steatotic liver disease; MRI, magnetic resonance imaging; US, ultrasound.

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Table 1.Populations at Risk for Hepatocellular Carcinoma in Which Surveillance Is Advised

Risk and advice	Patient population	Annual incidence
Sufficient risk; surveillance advised	Child-Pugh class A or B cirrhosis, any etiology	Varies, >1.0%
	Hepatitis C (viremic and post-SVR)	
	Hepatitis B	
	Alcohol-associated liver disease	
	Metabolic-associated steatotic liver disease	
	Other etiologies (eg, primary biliary cholangitis and autoimmune liver disease)	Varies, >1.0%
	Child-Pugh class C cirrhosis, if transplant candidate	
	Chronic hepatitis B without cirrhosis	
	Men aged >40 y from endemic countries	
	Women aged >50 y from endemic countries	
	Patients from Africa at an earlier age	
	Family history of HCC	
	PAGE-B score ≥10	
Insufficient risk to advise routine surveillance in the absence of new biomarkers and/or better risk stratification	Patients with stage F3 fibrosis or lower, including those with metabolic-associated liver disease and hepatitis C	<0.2%

Adapted from Singal et al.¹²
SVR, sustained virologic response.

transplantation derive little benefit due to competing mortality risks. However, for patients with Child-Turcotte-Pugh class C cirrhosis on transplant waitlists, surveillance remains important to identify early-stage tumors (which may increase transplant priority) and those with larger tumors that may preclude liver transplantation. Surveillance is not advised for patients with a life expectancy <1–2 years due to other comorbidities that cannot be remedied by means of liver transplantation or HCC therapies.

The strongest evidence supporting HCC surveillance comes from a randomized trial in patients with chronic HBV infection, which found that surveillance reduced HCC mortality (hazard ratio, 0.63; 95% CI, 0.41–0.98).¹³ Although there are no similar randomized trials for patients with cirrhosis, meta-analyses of observational studies demonstrate the benefits of surveillance, including improvements in early detection, access to curative treatment, and survival. Although these studies are subject to potential lead-time and length-time biases, survival benefits have persisted in analyses adjusting for these factors. Given the consistent directionality of evidence and improvements in downstream outcomes, the absence of randomized data should not be interpreted as lack of support for surveillance in appropriate high-risk populations.

Abdominal US has been the cornerstone of surveillance, although it is limited by operator dependence and worse

performance in patients with obesity, severe steatosis, or hepatic decompensation.¹⁴ The addition of AFP provides an incremental benefit in sensitivity (63% for US and AFP compared with 51% for US alone).¹⁵ Although this comes with a small decrease in specificity, the diagnostic odds ratio for the combination is higher than for US alone. Cost-effectiveness analyses also support the use of US and AFP in combination.¹⁶

Although computed tomography and magnetic resonance imaging (MRI) may offer higher sensitivity and specificity than US, they should not be considered for routine surveillance due to concerns related to radiation and contrast exposure, high cost, and low availability due to limited radiology capacity. Decision models suggest MRI-based surveillance may be cost-effective in select populations, but the incremental cost-effectiveness ratio varies widely based on HCC incidence, cirrhosis etiology, and test costs.¹⁷

HCC surveillance should be performed semiannually. This frequency detects tumors at an earlier stage and improves survival compared with annual surveillance.¹⁸ Despite this, surveillance remains significantly underused (estimated at approximately 24%) even among patients managed by subspecialists.¹⁹ Efforts to increase uptake include outreach (eg, mailed invitations) and “in-reach” (eg, electronic medical record reminders and provider education). These approaches have been found to significantly increase participation in surveillance programs.^{20,21}

Best Practice Advice 3: Among patients without cirrhosis, only a subset of those with chronic hepatitis B virus should undergo hepatocellular carcinoma surveillance; surveillance is not advised for those without cirrhosis from other etiologies due to the low annual incidence rate.

Data are evolving regarding the value of surveillance in patients with HCV cirrhosis who have achieved cure (ie, sustained virologic response), patients with cured HCV who do not have cirrhosis, and patients with MASLD who do not have cirrhosis. Recent modeling studies suggest surveillance is cost-effective when annual HCC incidence is approximately 1.0% per year.¹⁶ Based on available data, patients with HCV cirrhosis remain at increased HCC risk for up to 10 years after achieving sustained virologic response,²² so surveillance should continue indefinitely. In contrast, in patients post sustained virologic response who never had cirrhosis, HCC incidence is significantly lower than the cost-effectiveness threshold for surveillance.

Up to 20% of MASLD-HCC occurs in individuals without cirrhosis.²³ However, the overall HCC incidence in this group is low (~ 0.008 per 100 person-years).²⁴ More accurate tools are needed to risk stratify patients with cured HCV and MASLD without cirrhosis and avoid unnecessary screening.

Although routine HCC surveillance is not advised in patients with F3 fibrosis, clinicians should recognize the potential for misclassification of fibrosis stage using noninvasive testing. In select cases, such as patients with uncertainty regarding fibrosis staging or additional risk factors (eg, family history of HCC), individualized discussion and shared decision making may be appropriate.

Best Practice Advice 4: Although the benefits of hepatocellular carcinoma surveillance are well established, it is important for clinicians to consider the potential physical, psychological, and financial harms associated with the process.

Although current surveillance modalities are noninvasive, they are not without risks. For example, false-positive results detected on US may lead to unnecessary contrast-enhanced imaging or biopsy. Although serious complications are rare, physical harms of liver biopsy include bleeding, infection, and, rarely, tumor seeding. In 1 study of 614 patients enrolled in HCC surveillance, 19.2% ($n = 118$) had abnormal screening results, but only 26 developed HCC. Physical harms were observed in 10% of patients enrolled, but fortunately most were mild or moderate in severity.²⁵

Potential psychological harms are often overlooked.²⁶ Uncertainty from indeterminate lesions or other incidental findings may cause anxiety, stress, and diminished quality of life. Even benign findings may prompt months of further testing and patient worry.

Financial harms are also an important concern. Although US is relatively low-cost, expenses related to repeat testing and follow-up diagnostic imaging can accumulate. In 1 trial of HCC screening outreach, 21.8% of patients reported moderate to severe financial burden from surveillance.²⁶ Missed work and travel costs add to the cost burden.

Given these potential harms, clinicians should individualize surveillance decisions. For patients with limited life

expectancy and those who are not transplant candidates, surveillance offers little benefit and may only expose them to harm. Shared decision making is key; patients should understand the limitations of surveillance, including the potential for false-positive and false-negative results, emotional distress, and financial implications.

Best Practice Advice 5: Several novel blood-based biomarkers, such as GALAD, and radiologic biomarkers are undergoing clinical validation. Some are already commercially available, but evidence is insufficient to support their use in routine hepatocellular carcinoma surveillance. These assays should not replace guideline-recommended tests, although their accessibility and potential cost-effectiveness may support broader use once sufficiently validated.

The limited sensitivity of US with AFP has prompted evaluation of novel surveillance strategies¹⁵ broadly categorized as blood-based biomarkers and radiologic approaches. Blood-based biomarkers are attractive due to ease of use and the potential to improve surveillance uptake.¹⁹ Even modest improvements in test sensitivity may translate into meaningful gains in early detection at a population level.²⁷

Among blood-based approaches, lectin-reactive fraction of AFP and des- γ -carboxyprothrombin have been studied most extensively. Although these biomarkers do not outperform AFP individually, combination strategies have led to the development of composite scores such as GALAD and the Hepatocellular Carcinoma Early Detection Screening algorithm.²⁸ The GALAD score incorporates age, sex, AFP, lectin-reactive fraction of AFP, and des- γ -carboxyprothrombin to estimate HCC risk.²⁹ In case-control studies, GALAD demonstrated sensitivities of 72%–82% for early-stage HCC³⁰; however, performance has been more modest in longitudinal cohorts. In a phase 3 validation study of 1558 patients with cirrhosis, GALAD (cutoff of -1.36 , assessed 6 months before HCC diagnosis) achieved 65% sensitivity and 82% specificity.

These cutoffs are typically benchmarked at approximately 80% specificity, corresponding to a $\sim 20\%$ false-positive rate, to prioritize early-stage detection. In cancer surveillance settings, greater sensitivity may be favored to identify potentially curative disease, and available data suggest that patients and other stakeholders often prioritize the benefits of surveillance over potential harms.^{31,32} Nevertheless, the psychological and financial consequences of false-positive testing results remain incompletely defined.

Few observational studies suggest that using blood-based biomarkers as an adjunct to imaging may improve overall surveillance sensitivity. However, this strategy increases cost and does not address patient-level barriers to imaging-based screening. Importantly, no prospective studies have directly compared an adjunct biomarker-plus-imaging approach to either modality alone. Therefore, based on currently available evidence, the use of blood-based biomarkers as an adjunct (rather than replacement) to imaging cannot be recommended at this time.

The Hepatocellular Carcinoma Early Detection Screening algorithm, version 2.0, which incorporates AFP trends along with lectin-reactive fraction of AFP, des- γ -carboxyprothrombin, ALT, platelet count, age, and cirrhosis

etiology, demonstrated improved overall performance compared with other scores (eg, GALAD and ASAP) in phase 3 evaluation; however, sensitivity for early-stage HCC remained limited (47%) and optimal cutoff values require further validation.

Emerging liquid biopsy approaches, including cell-free DNA methylation assays, circulating tumor cells, and extracellular vesicle profiling, have had promising early-phase results, with some studies reporting sensitivities and specificities >80%. However, most data derive from phase 2 studies, few have received US Food and Drug Administration breakthrough device designation, and are not yet recommended for use in clinical practice.³³

Ongoing prospective trials, including the TRACER trial,³⁴ will provide more definitive data regarding the effectiveness and potential harms of biomarker-based surveillance. At present, although promising, novel blood-based and liquid biopsy approaches require further validation before routine clinical implementation.

There is also growing interest in novel radiologic approaches for early detection.³⁵ Abbreviated MRI, which reduces the number of sequences, examination time, and costs, has shown early promise.³⁶ In a prospective study of 208 patients, annual noncontrast abbreviated MRI had increased sensitivity for early-stage HCC compared with semiannual US (71% and 45%, respectively).³⁷ Small cohort studies also suggest a performance comparable with conventional MRI.³⁸ The PREMIUM (Preventing Liver Cancer Mortality Through Imaging With Ultrasound vs MRI) trial is evaluating abbreviated MRI in a large United States-based study and will provide definitive data on its utility for HCC surveillance.

Best Practice Advice 6: Multicancer blood-based biomarker detection panels should not be used in the screening or surveillance of patients at risk for hepatocellular carcinoma.

An early retrospective study found that 62% of localized tumors (eg, colorectal, breast, lung, and ovarian) were detected using targeted error-correction sequencing of cell-free DNA.³⁹ Panels combining cell-free DNA mutations with protein biomarkers (eg, CancerSEEK) have further improved sensitivity for detecting cancers such as ovary, liver, stomach, pancreas, and esophagus.⁴⁰

In a prospective case-control study of 2800 participants with various tumor types and stages, whole-genome DNA methylation achieved 98% specificity and 34%–39% sensitivity, showing promise for cancer detection. Despite these advances, multidetection panels have been studied in general populations with few patients with cirrhosis or HCC. Therefore, they should not be used as standard tools for HCC surveillance.

Best Practice Advice 7: Many hepatocellular carcinoma risk-stratification scores exist for patients with cirrhosis. However, few scores have undergone sufficient validation to support their use in clinical practice.

HCC risk varies among patients with cirrhosis; with annual incidence rates ranging from 1% to 8%.⁴¹ By improving risk-stratification tools, surveillance intensity and test modality could be tailored precisely based on a

patient's individual risk, enabling resource-intensive efforts to be directed toward those at highest risk and reducing screening intensity and related harms for lower-risk patients.⁴²

Current HCC risk-stratification models offer some utility but face limitations and have been covered in recent reviews (selected subset in Table 2^{43–47} published since the review by Singal et al⁴⁸). Most were derived from cohorts of patients with viral etiologies and limited diversity, including few women and racial and ethnic minority groups. A recent study conducted in a contemporary cohort of patients with cirrhosis with racial, ethnic, and etiologic diversity, developed an HCC risk-stratification index addressing some limitations of previous models.⁴⁶

Most existing models combine demographic and clinical factors. This approach has the practical advantage of relying on routinely collected data, resulting in relatively simple tools that could be implemented in clinical practice. Although these basic models provide a good foundation for predicting HCC risk in cirrhosis, their performance is modest (model discrimination estimates from ~0.70 to 0.80), precluding widespread clinical use. Other lifestyle, genetic, and metabolic factors are associated with HCC risk.^{49–58}

Compared with simple models, multidimensional models incorporating factors from these domains (ie, genetic, metabolic, and radiomic markers) may better distinguish low-risk from high-risk patients. In a study of patients in prospective cohorts, a 7-single nucleotide polymorphism genetic risk score was associated with HCC risk. Adding this genetic risk score to a clinical model improved the clinical model's performance (C-indices of 0.786 and 0.783).⁴⁵ Similarly, El-Serag et al⁴⁷ derived and internally validated a 9-biomarker signature that predicted HCC risk in cirrhosis. This biomarker panel resulted in a small but statistically significant improvement in predictive value for future HCC risk compared with a clinical model derived from demographic, clinical, and lifestyle variables.

Although building on simple clinical models by incorporating factors from multiple domains may improve their performance, the added costs of additional testing must be weighed against potential incremental improvements in performance when considering the overall value of a multidimensional model. Most models have not undergone extensive external validation to support their broad applicability. For a model to be relevant, clinicians should be able to use the predicted risk to guide decisions about surveillance or risk reduction. However, all current HCC risk-stratification models produce false-positive and false-negative results. There is no consensus on acceptable levels of false-negative or false-positive results, and the widespread use of risk-stratification models will remain limited until clinically relevant thresholds and corresponding decisions are agreed on.

Finally, there is a lack of data on the impact of risk-stratification models and decision rules on clinician behavior, patient behavior, and subsequent patient outcomes. A randomized controlled trial to examine the

Table 2. Select Hepatocellular Carcinoma Risk Stratification Models in Patients With Cirrhosis Published Since Recent Reviews

First Author, Year	Study design	Etiology	Cirrhosis	Variables	Output	HCC risk	Group with zero HCC risk	External validation
Johnson, 2022 ⁴³	Retrospective cohort	55% HCV (50% SVR at the end of follow-up), 24% HBV, 21% other etiologies	25% cirrhosis	aMAP score, based on age, gender, platelet count and the ALBI score	C-index, 0.78 (95% CI, 0.76–0.80) overall	Incidence rates were 0.98, 7.05, and 29.1 events per 1000-person years for individuals in the low-risk, moderate-risk, and high risk aMAP groups, respectively	No	Yes
Ioannou, 2019 ⁴⁴	Retrospective cohort	Alcohol-related or MASLD-related cirrhosis	100% cirrhosis	Age, gender, diabetes, body mass index, platelet count, serum albumin, and aspartate aminotransferase to alanine aminotransferase ratio	AUROC, 0.75 for MASLD-cirrhosis and 0.76 for ALD cirrhosis	3 groups with progressively higher annual HCC incidence: high risk >3%, medium risk 1%–3%, low risk <1%	No	No
Nahon, 2023 ⁴⁵	Retrospective cohort	Alcohol-related and/or HCV-cured cirrhosis	100% cirrhosis	PNPLA3, TM6SF2, HSD17B13, APOE, MBOAT7, and WNT3A-WNT9A rs708113 with and without clinical factors (age, sex, diabetes, platelet count, GGT, albumin)	C-Index of clinical model 0.769 for 5-y HCC risk Addition of GRS increased to 0.786	3 groups with progressively increased 5-y HCC incidence (group 1 (n = 627 [5.4%]), group 2 (n = 276 [10.7%]), and group 3 (n = 242 [15.3%]))	No	NA
Kanwal, 2023 ⁴⁶	Prospective cohort	Multi-etiology	100% cirrhosis	Age, sex, smoking, alcohol use, body mass index, etiology, AFP, albumin, alanine aminotransferase, and platelet levels	AUROC, 0.77 (95% CI, 0.71–0.83) at 2 y	3-y HCC risk increased per decile, lowest decile 0.4% to highest decile at 18.7%	No	Yes
El-Serag, 2024 ⁴⁷	Prospective cohort	Multi-etiology	100% cirrhosis	Model based on clinical factors (Kanwal et al ⁴⁴) with a 9-marker panel	AUROC, 0.78 (95% CI, 0.77–0.78) at 1 y and 0.748 (95% CI, 0.74–0.75)	NA	NA	No

ALBI, albumin–bilirubin; ALD, alcohol-associated liver disease; AUROC, area under the receiver operating characteristic curve; GGT, γ -glutamyl transferase; GRS, genetic risk score; NA, not applicable.

Table 3. Externally Validated Risk Stratification Models for Patients With Chronic Hepatitis B Virus Without Cirrhosis

Model	Components	Risk groups and cutoffs
PAGE-B	Age, sex, platelet count	Low risk: <10 Intermediate risk: 10–17 High risk: ≥18
REAL-B	Age, male, alcohol use, cirrhosis, diabetes, platelet, AFP	Low risk: <4 Intermediate risk: 4–7 High risk: 8–13
AASL-HCC	Age, male, albumin, cirrhosis	Low risk: ≤5 Intermediate risk: 6–19 High risk: ≥20

clinical utility of HCC risk stratification in patients with cirrhosis may not be forthcoming. A decision-analytical approach could help bridge this gap by simulating the clinical course of cirrhosis and quantifying how different risk stratification models may reduce HCC mortality through improved surveillance and early detection.

Best Practice Advice 8: Among patients with chronic hepatitis B virus infection without cirrhosis, PAGE-B and REAL-B scores can stratify patients based on their future risk of hepatocellular carcinoma.

Several risk prediction models have been developed to estimate future HCC risk in patients with HBV, enabling risk-stratified surveillance and individualized clinical decision making. Examples include REACH-B (includes age, sex, alanine aminotransferase, hepatitis B e-antigen, and HBV DNA)⁵⁹; PAGE-B (includes age, sex, and platelet)⁶⁰; mPAGE-B (includes age, sex, platelet, and albumin)⁶¹; CU-HCC (includes age, cirrhosis, albumin, bilirubin, and HBV DNA)⁶²; HCC-RESCUE (includes age, sex, and cirrhosis)⁶³; CAMD (includes cirrhosis, age, sex, and diabetes mellitus)⁶⁴; APA-B (includes age, platelet, and AFP)⁶⁵; REAL-B (includes age, sex, alcohol use, cirrhosis, diabetes mellitus, platelet, and AFP)⁶⁶; AASL-HCC (includes age, albumin, sex, and liver cirrhosis)⁶⁷; and RWS-HCC (includes age, sex, cirrhosis, and AFP)⁶⁸ scores.

In a recent comparative study,⁶⁹ most models showed good overall discrimination, with area under the curve values ≥0.75 for 7 of the 10 models (ie, CAMD, APA-B, HCC-RESCUE, CU-HCC, AASL-HCC, RWS-HCC, and REAL-B). Area under the curve values for 3 models that incorporated AFP (ie, APA-B, RWS-HCC, and REAL-B) were the highest (all, >0.80). Three models identified low-risk groups with minimal misclassification. None of the patients in the low-risk groups defined by PAGE-B, mPAGE-B, AASL-HCC, or REAL-B developed HCC during the study period. However, REAL-B and AASL-HCC demonstrated better discrimination than PAGE-B in this cohort. Collectively, these data suggest that PAGE-B, AASL-HCC, and REAL-B can be used to risk stratify patients with HBV. Table 3 includes the components and cutoffs for these models, which can be applied to risk stratify patients with HBV. Other factors, including whether the patient is from an endemic region, age, or

family history of HCC should also be considered in making surveillance decisions.

In the future, we expect to see new (and iterative) models and approaches for HCC risk stratification in HBV. For example, a recent study incorporated HBV DNA viral load into the PAGE-B model, creating the PAGED-B model, which performed well in patients who underwent a phase change from chronic hepatitis B to hepatitis B e-antigen-positive HBV.⁷⁰ Artificial intelligence has emerged as a powerful tool to improve HCC risk prediction. For example, a machine learning approach was used to derive the SMART-HCC score, which includes a liver stiffness measurement and 8 other parameters, including cirrhosis etiology, age, alanine aminotransferase, alkaline phosphatase, creatinine, platelet count, sex, and hypertension.⁷¹ Although promising, these models will require further external validation before their widespread use can be recommended.

Conclusions

Although several novel biomarkers have shown potential in phase 2 and phase 3 validation studies, US combined with AFP remains the recommended HCC surveillance strategy in guidelines. Ongoing phase 4 and 5 biomarker clinical utility trials, such as TRACER and PREMIUM, will provide data to guide the integration of novel biomarkers into clinical practice. To optimize surveillance effectiveness and minimize harms, there is an unmet need for improved risk stratification to identify both high-risk individuals more accurately and to lower-risk individuals who may be monitored or surveilled with less intensity. This collaborative process will enable earlier adoption of precision screening strategies in clinical practice, ultimately improving HCC early detection and outcomes.

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Conflicts of interest

These authors disclose the following: Nicole E. Rich has served as consultant or on advisory boards for Genentech, Eisai, Exelixis, and AstraZeneca. Augusto Villanueva has received consulting fees from FirstWorld, Pioneering Medicine, and Genentech; advisory board fees from BMS, Roche, AstraZeneca, Eisai, and NGM Pharmaceuticals; and research support from Eisai; he has stock options from Espervita and Atzeye and is listed as an inventor on a patent related to early detection of hepatocellular carcinoma (PCT/US20/61441). Jorge A. Marrero is a consultant for Glycotest and a member of the data safety monitoring board for 2 studies sponsored by AstraZeneca. The remaining author discloses no conflicts.

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